

MARKET EQUILIBRIUM

When quantity demanded and quantity supplied are equal to each other, it is called market equilibrium.

Example 1:

(i) $Q_d = 10 - 2p$

(ii) $Q_s = 4 + p$

Market equilibrium is achieved where

$$Q_d = Q_s$$

$$10 - 2p = 4 + p$$

$$-p - 2p = 4 - 10$$

$$-3p = -6$$

$$P = \frac{-6}{-3}$$

$$P = 2$$

Equilibrium price is entered in Demand function.

$$Q_d = 10 - 2p$$

$$Q_d = 10 - 2(2)$$

$$Q_d = 10 - 4$$

$$Q_d = 6$$

Equilibrium Price is entered in Supply Function.

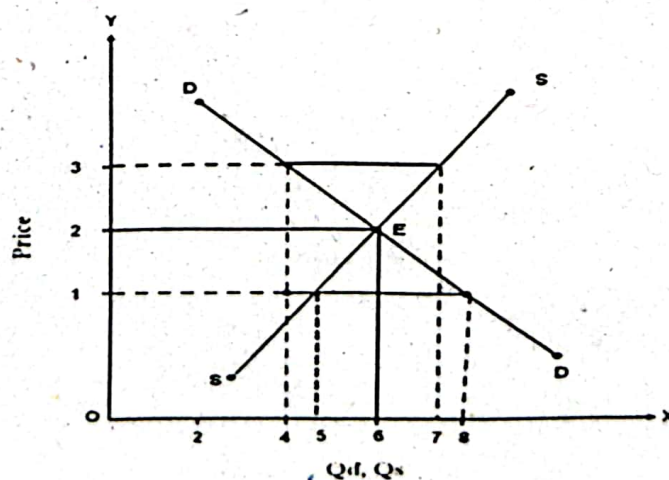
$$Q_s = 4 + (2)$$

$$Q_s = 6$$

$$Q_d = Q_s$$

$$6 = 6$$

Price	$Q_s = 4 + p$	$Q_d = 10 - 2p$
1	$Q_s = 4 + p$ $= 4 + 1 = 5$	$Q_d = 10 - 2(1)$ $= 10 - 2 = 8$
2	$Q_s = 4 + p$ $= 4 + 2 = 6$	$Q_d = 10 - 2(2)$ $= 10 - 4 = 6$
3	$Q_s = 4 + p$ $= 4 + 3 = 7$	$Q_d = 10 - 2(3)$ $= 10 - 6 = 4$



According to Diagram, equilibrium quantity is 6 while equilibrium price is 2.

$$Q_d = -20 + 3p$$

$$Q_s = 220 - 5p$$

$$Q_d = Q_s$$

$$220 - 5p = -20 - 220$$

$$-8p = -240$$

$$\bar{P} = \frac{240}{8} = 30$$

$$\bar{Q} = -20 + 3(30)$$

$$\bar{Q} = -20 + 90 = 70$$

$$\bar{P} = 30$$

$$Q = 70$$

$$\bar{Q} = 220 - 5(30)$$

$$\bar{Q} = 220 - 150$$

$$Q = 70$$

Example 3:

$$P + Q^2 + 3Q - 20 = 0$$

Supply Function

$$P - 3Q^2 + 10Q = 5$$

$$P - 3Q^2 + 10Q = 5$$

$$P = 3Q^2 - 10Q + 5$$

Substituting $P = 3Q^2 - 10Q + 5$ in demand function

$$(3Q^2 - 10Q + 5) + Q^2 + 3Q - 20 = 0$$

$$4Q^2 - 7Q - 15 = 0$$

$$a = 4, b = -7, c = -15$$

$$Q = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4)(-15)}}{2(4)} = \frac{7 \pm \sqrt{49 + 240}}{8} = \frac{7 \pm \sqrt{289}}{8}$$

$$= \frac{7+17}{8} = \frac{24}{8} = 3$$

$$= \frac{7-17}{8} = \frac{-10}{8} = -1.25$$

by putting $q=3$ in any equation.

$$P + (3)^2 + 3(3) - 20 = 0$$

$$P = 2$$

EXERCISE 5

i) $Q_s = -20 + 3P$

$$Q_d = 220 - 5P$$

iii) $13P - Q_s = 27$

$$Q_d + 4P - 24 = 0$$

v) $P + Q^2 + 3Q - 20 = 0$

$$P - 3Q^2 + 10Q = 5$$

ii) $Q_s + 32 - 7P = 0$

$$Q_d - 128 + 9P = 0$$

iv) $3P + Q^2 + 5Q - 102 = 0$

$$P - 2Q^2 + 3Q = 71$$

Answers

i) $P = 30$

$$Q = 70$$

iii) $P = 3$

$$Q = 12$$

v) $P = 2$

$$Q = 3$$

ii) $P = 10$

$$Q = 38$$

iv) $P = 6$

$$Q = 7$$